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In [17]: import pandas as pd
df=pd.read_excel('breast cancer patients.xlsx')
print('Table1 breast cancer patients:')
print(df)
print('-----')
from lifelines import KaplanMeierFitter
import matplotlib.pyplot as plt
kmf=KaplanMeierFitter()
print('Overall survival curve:')
kmf.fit(
    durations=df['Survival_months'],
    event_observed=df['Status']
)
kmf.plot(
    show_censors=True
)
plt.title('Overall survival curve')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.savefig(
    'Overall survival curve of breast cancer patients.png',
    dpi=300
)
plt.show()
print('-----')
print('Survival curves by stage:')
stages=['I','II','III']
for stage in stages:
    stage_data=df[df['Stage']==stage]
    kmf.fit(
        durations=stage_data['Survival_months'],
        event_observed=stage_data['Status'],
        label='stage'+stage
    )
    kmf.plot(
        show_censors=True
    )
plt.title('Survival curves by stage')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by stage.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
from lifelines.statistics import multivariate_logrank_test
result1=multivariate_logrank_test(
    df['Survival_months'],
    df['Stage'],
    df['Status']
)
print('p value:',result1.p_value)
print('因为P值大于0.05，所以分期的生存曲线没有统计学意义。')
print('-----')
print('Survival curves by ER:')
for group in ['-','+']:

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ix=df['ER']==group
kmf.fit(
    durations=df.loc[ix,'Survival_months'],
    event_observed=df.loc[ix,'Status'],
    label='ER'+group
)
kmf.plot(
    show_censors=True
)
plt.title('Survival curves by ER')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by ER.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
from lifelines.statistics import logrank_test
group0=df[df['ER']=='-']
group1=df[df['ER']=='+']
result2=logrank_test(
    group0['Survival_months'],
    group1['Survival_months'],
    event_observed_A=group0['Status'],
    event_observed_B=group1['Status']
)
print('p value:',result2.p_value)
print('因为P值大于0.05，所以关于ER的生存曲线没有统计学意义。')
print('-----')
print('Survival curves by Ki67:')
df['Ki67_group']=df['Ki67'].apply(lambda x:'high' if x>='30%' else 'low')
for kgroup in ['high','low']:
    ix=df['Ki67_group']==kgroup
    kmf.fit(
        durations=df.loc[ix,'Survival_months'],
        event_observed=df.loc[ix,'Status'],
        label='Ki67'+kgroup
    )
    kmf.plot(
        show_censors=True
    )
plt.title('Survival curves by Ki67')
plt.xlabel('Time(months)')
plt.ylabel('Survival probability')
plt.legend()
plt.savefig(
    'Survival curves by Ki67.png',
    dpi=300,
    bbox_inches='tight'
)
plt.show()
Kgroup0=df[df['Ki67_group']=='low']
Kgroup1=df[df['Ki67_group']=='high']
result2=logrank_test(
    Kgroup0['Survival_months'],
    Kgroup1['Survival_months'],
    event_observed_A=Kgroup0['Status'],
    event_observed_B=Kgroup1['Status']
)

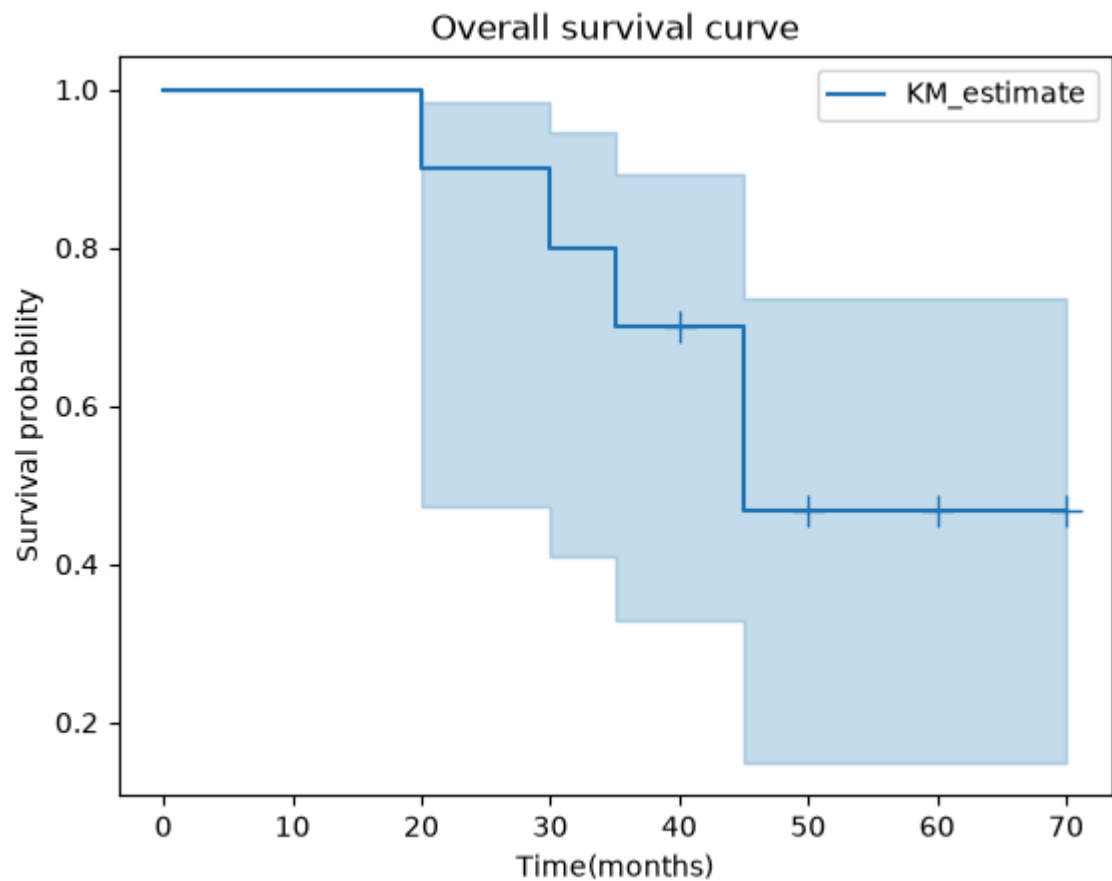
```

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)  
print('P value:',result2.p_value)  
print('因为P值大于0.05，所以Ki67的生存曲线没有统计学意义。')
```

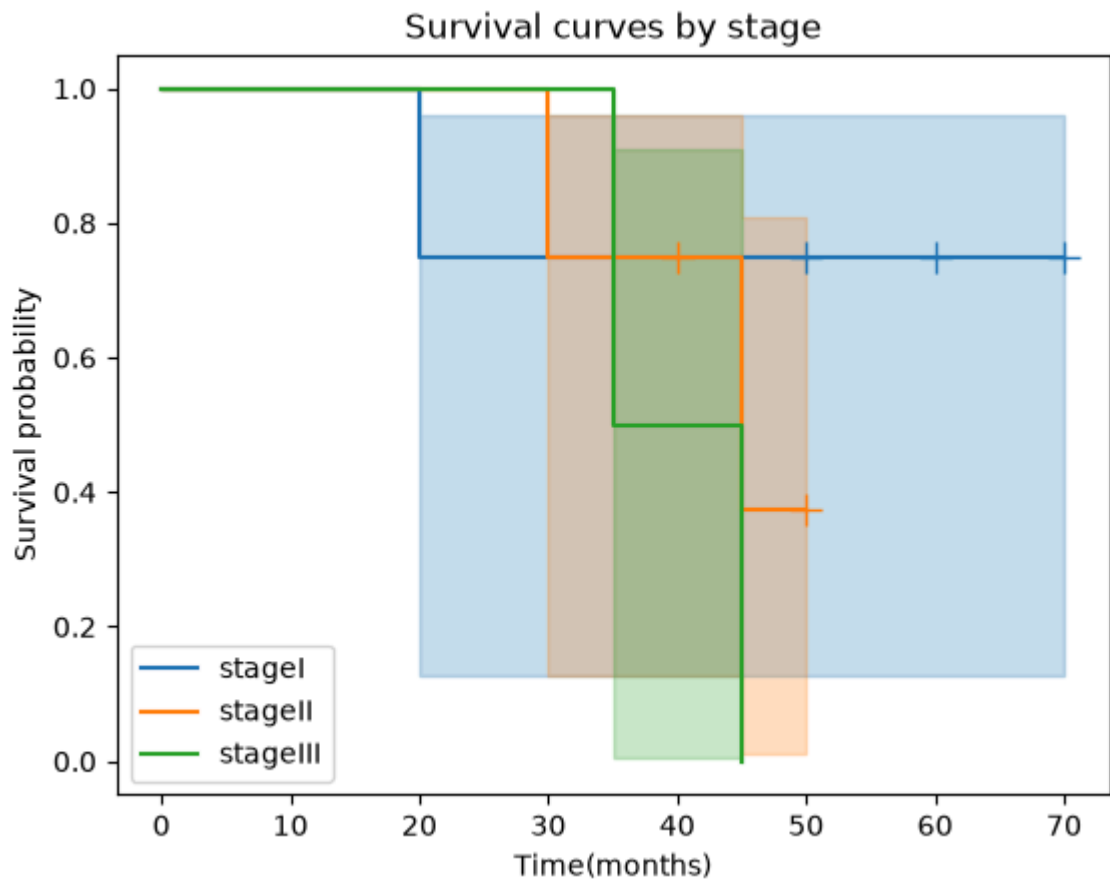
Table1 breast cancer patients:

	Unnamed: 0	Patient_ID	Age	Stage	ER	PR	HER2	Ki67	Survival_months	Status
0	0	P001	45	I	+	+	-	10%	70	0
1	1	P002	55	II	+	-	-	40%	50	0
2	2	P003	62	II	-	-	+	70%	45	1
3	3	P004	38	I	+	+	-	20%	50	0
4	4	P005	42	II	+	-	-	30%	30	1
5	5	P006	57	III	-	+	+	65%	45	1
6	6	P007	60	III	-	+	+	75%	35	1
7	7	P008	49	I	+	+	-	50%	20	1
8	8	P009	51	II	+	-	-	45%	40	0
9	9	P010	39	I	+	+	-	30%	60	0

Overall survival curve:



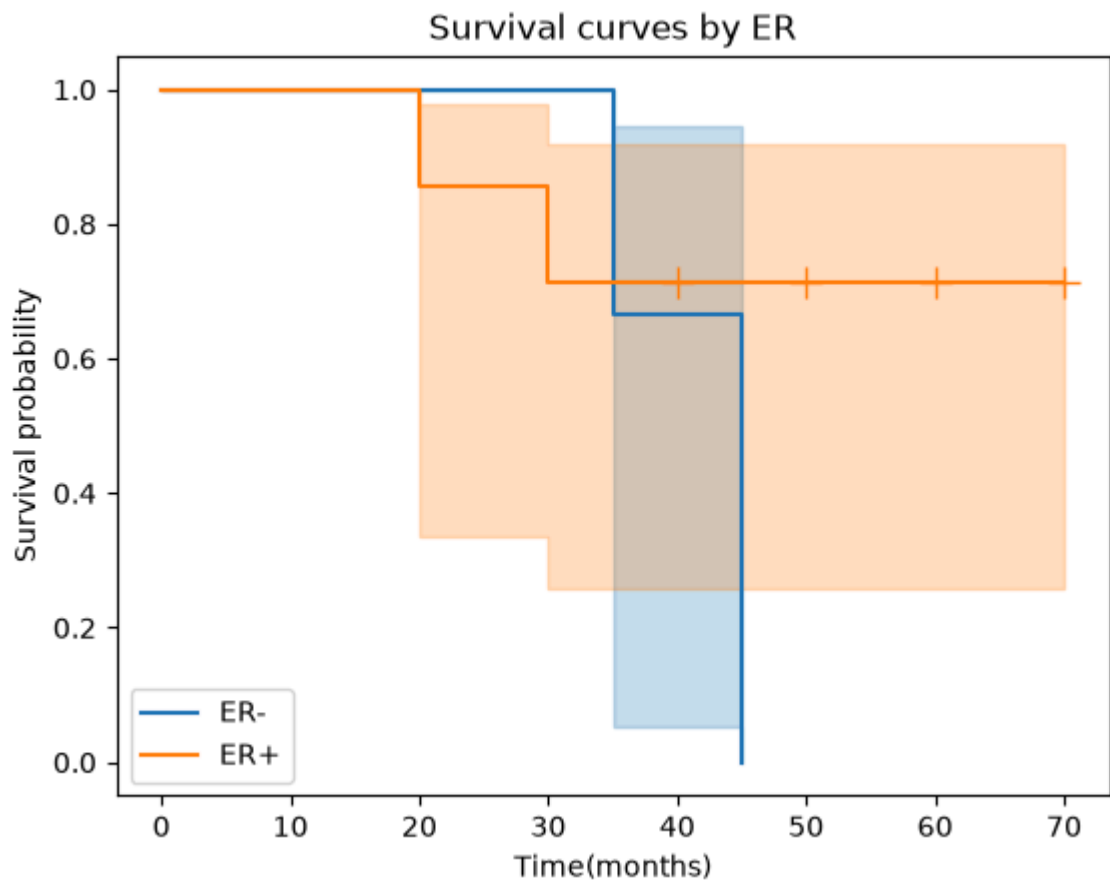
Survival curves by stage:



p value: 0.42136039096062616

因为P值大于0.05，所以分期的生存曲线没有统计学意义。

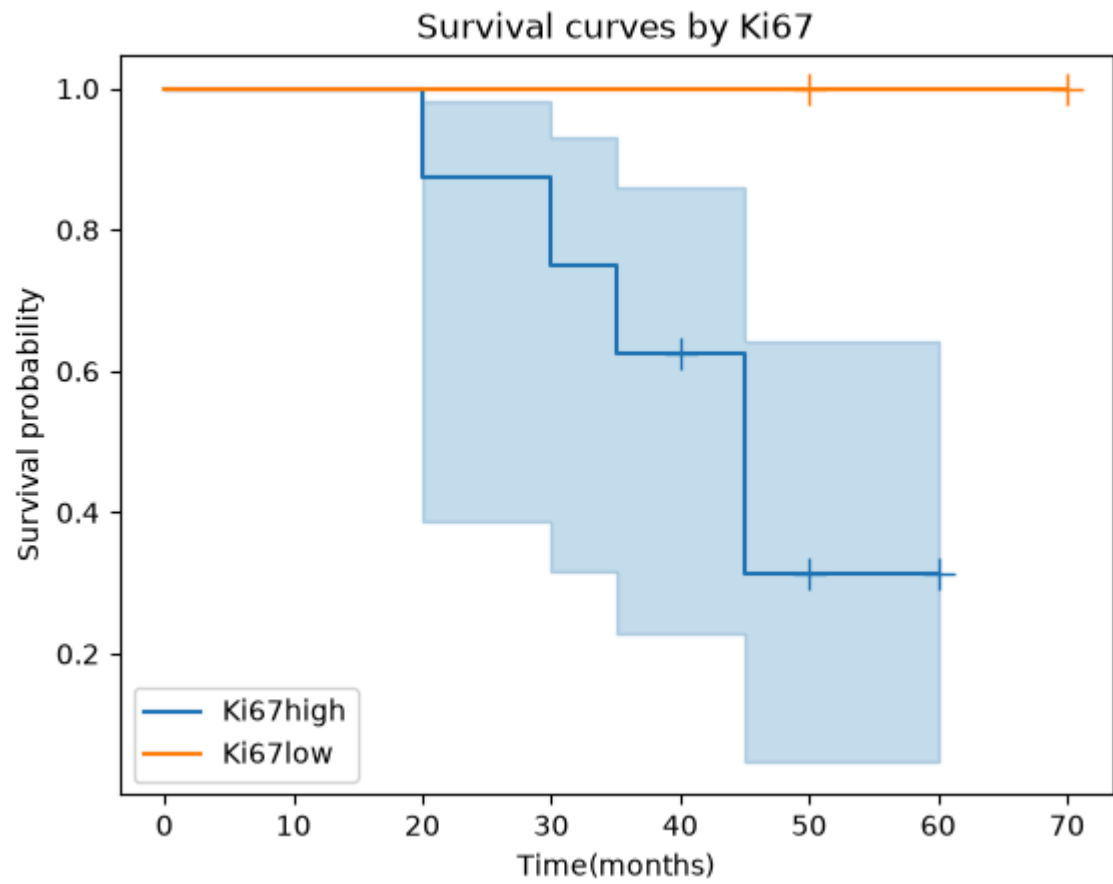
Survival curves by ER:



p value: 0.19000537623733096

因为P值大于0.05，所以关于ER的生存曲线没有统计学意义。

Survival curves by Ki67:



P value: 0.15254462500774932

因为P值大于0.05，所以Ki67的生存曲线没有统计学意义。

In []: